Time Dependent Hartree-Fock and Linear Response Theory

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Abstract

This note reviews time dependent Hartree-Fock theory and how can it be used to calculate the linear response of a many-body system to an external perturbation. We specifically demonstrate that the TDHF response function has corrections from exchange interactions compared to the random phase approximation (RPA) response function.

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1 Density Matrix Formalism of EoM

We start with a general many-body Hamiltonian [1]:

$$\hat{H} = \sum_{ij} [t_{ij} + o_{ij} f(t)] \hat{c}_i^\dagger \hat{c}_j + \frac{1}{2} \sum_{ijkl} V_{ijkl} \hat{c}_i^\dagger \hat{c}_j^\dagger \hat{c}_k \hat{c}_l \tag{1}$$

where $o_{ij} f(t)$ is the external field perturbation. The Hemitinity requires:

$$t_{ij} = t_{ji}^*, \quad o_{ij} = o_{ji}^*, \quad V_{ijkl} = V_{klij}^* \tag{2}$$

Femionic anti-commutation requires:

$$V_{ijkl} = -V_{jikl} = -V_{ijlk} = V_{jilk} \tag{3}$$

The single-particle density matrix (time-dependent) is defined as:

$$\rho_{mn} = \langle \hat{c}_n^\dagger \hat{c}_m \rangle = \langle \Psi_0 | \, e^{iHt/\hbar} \hat{c}_n^\dagger \hat{c}_m e^{-iHt/\hbar} \, | \Psi_0 \rangle \tag{4}$$

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The equation of motion is derived as:

$$
\begin{aligned}
i\hbar\partial_t\rho_{mn} =& \langle[\hat{c}_n^\dagger\hat{c}_m, \hat{H}]\rangle && (5)\\
=& \langle\hat{c}_n^\dagger[\hat{c}_m, \hat{H}]\rangle - \langle[\hat{H}, \hat{c}_n^\dagger]\hat{c}_m\rangle && (6)\\
=& (t_{mj} + o_{mj}f(t))\rho_{jn} - (t_{in} + o_{in}f(t))\rho_{mi} + V_{mikl}\langle\hat{c}_n^\dagger\hat{c}_i^\dagger\hat{c}_k\hat{c}_l\rangle - V_{klni}\langle\hat{c}_l^\dagger\hat{c}_k^\dagger\hat{c}_i\hat{c}_m\rangle, && (7)
\end{aligned}
$$

where the Einstein summation is implied. We list the commutator involving interaction term for reference:

$$
\begin{aligned}
[\hat{c}_m, \hat{V}] =& \frac{1}{2}\sum_{ijkl} V_{ijkl}[\hat{c}_m, \hat{c}_i^\dagger\hat{c}_j^\dagger\hat{c}_k\hat{c}_l] && (8)\\
=& \frac{1}{2}\sum_{ijkl} V_{ijkl}\left(\hat{c}_i^\dagger\hat{c}_j^\dagger[\hat{c}_m, \hat{c}_k\hat{c}_l] + [\hat{c}_m, \hat{c}_i^\dagger\hat{c}_j^\dagger]\hat{c}_k\hat{c}_l\right) && (9)\\
=& \frac{1}{2}\sum_{ijkl} V_{ijkl}[\hat{c}_m, \hat{c}_i^\dagger\hat{c}_j^\dagger]\hat{c}_k\hat{c}_l && (10)\\
=& \frac{1}{2}\sum_{jkl} V_{mjkl}\hat{c}_j^\dagger\hat{c}_k\hat{c}_l - \frac{1}{2}\sum_{ikl} V_{imkl}\hat{c}_i^\dagger\hat{c}_k\hat{c}_l && (11)\\
=& \sum_{ikl} V_{mikl}\hat{c}_i^\dagger\hat{c}_k\hat{c}_l && (12)
\end{aligned}
$$

$$
[\hat{H}, \hat{c}_n^\dagger] = \left([\hat{c}_n, \hat{H}]\right)^\dagger \tag{13}
$$

2 Time Dependent Hartree-Fock

Under Hartree-Fock approximation, the ground state is assumed to be a slater determinant, therefore Eq. (7) can be written in a self-consistent mean-field of ρ:

$$
\begin{aligned}
i\hbar\partial_t\rho_{mn} =& (t_{mj} + o_{mj}f(t))\rho_{jn} - (t_{in} + o_{in}f(t))\rho_{mi} + V_{mikl}(\rho_{ln}\rho_{ki} - \rho_{kn}\rho_{li}) - V_{klni}(\rho_{ml}\rho_{ik} - \rho_{il}\rho_{mk})\\
=& [h(\rho) + of(t), \rho] && (14)\\
h(\rho)_{mn} :=& t_{mn} + \sum_{ij} V_{mijn}\rho_{ji} - V_{minj}\rho_{ji} && (15)
\end{aligned}
$$

Assuming the solution ρ is unique, we can prove $\rho^2 = \rho$:

$$
i\hbar\partial_t(\rho^2) = \rho[h + of(t), \rho] + [h + of(t), \rho]\rho = [h + of(t), \rho^2] \Rightarrow \rho^2 = \rho \tag{16}
$$

When the external field is absent ($f(t) = 0$), the system is in static ground state and ρ is time-independent from the definition (4), thus the mean-field EoM Eq. (14) reduces to:

$$
0 = [h(\rho^0), \rho^0] \tag{17}
$$

While the mean-field eigenequation is ($f(t) = 0$):

$$
\langle\hat{H}\rangle = t_{ij}\rho_{ji}^0 + \frac{1}{2}\sum_{ijkl} V_{ijkl}(\rho_{li}^0\rho_{kj}^0 - \rho_{ki}^0\rho_{lj}^0) \tag{18}
$$

$$
\frac{\delta\langle\hat{H}\rangle}{\delta\langle\Psi_0|} = E^0|\Psi_0\rangle \Rightarrow \sum_{ij} h(\rho^0)_{ij}\hat{c}_i^\dagger\hat{c}_j|\Psi_0\rangle = E^0|\Psi_0\rangle \tag{19}
$$

which solves ρ^0 as:

$$\rho^0_{\alpha\beta} = \sum_v \langle\alpha|v\rangle \langle v|\beta\rangle\,, \quad \text{where } |v\rangle \text{ is the occupied single-particle eigen state} \tag{20}$$

Just as ρ, ρ^0 is idempotent as well. ρ^0 acts like a projector onto the occupied subspace, conjugating with $\sigma^0 := \mathbf{1} - \rho^0$. This property becomes useful when we try to solve the linear response of ρ from Eq. (14). First we rewrite ρ as a perturbation series in order of $f(t)$:

$$\rho = \sum_{n=0} \rho^{(n)} \tag{21}$$

and the linear response of Eq. (7):

$$i\hbar\partial_t\rho^{(1)} = [h(\rho^0), \rho^{(1)}] + [h'(\rho^0)\cdot\rho^{(1)}, \rho^0] + [of(t), \rho^0]. \tag{22}$$

Up to first order, the idempotent property $\rho^2 = \rho$ implies:

$$\rho^0\rho^{(1)} + \rho^{(1)}\rho^0 = \rho^{(1)} \tag{23}$$

$$\Rightarrow \rho^{(1)}_{VV} = \rho^0\rho^{(1)}\rho^0 = 0, \quad \rho^{(1)}_{CC} = \sigma^0\rho^{(1)}\sigma^0 = 0 \tag{24}$$

which tells us that the linear response $\rho^{(1)}$ only has matrix elements between valence and conduction subspaces. We can project Eq. (22) onto the VC and CV subspaces to get a simplified EoM (the lower case $v, v', \ldots$ and $c, c', \ldots$ denote valence and conduction eigenstates of $h(\rho^0)$ respectively):

$$\begin{aligned} i\hbar\partial_t\rho^{(1)}_{vc} =& h(\rho^0)_{vv'}\rho^{(1)}_{v'c} - \rho^{(1)}_{vc'}h(\rho^0)_{c'c} - (h'_{vc,v'c'}\rho^{(1)}_{v'c'} + h'_{vc,c'v'}\rho^{(1)}_{c'v'}) - o_{vc}f(t) && (25)\\ =& \left[(\xi_v - \xi_c)\delta_{vv'}\delta_{cc'} - V_{vc'v'c} + V_{vc'cv'}\right]\rho^{(1)}_{v'c'} - \left(V_{vv'c'c} - V_{vv'cc'}\right)\rho^{(1)}_{c'v'} - o_{vc}f(t) && (26)\end{aligned}$$

$$\begin{aligned} i\hbar\partial_t\rho^{(1)}_{cv} =& \xi_c\rho^{(1)}_{cv} - \rho^{(1)}_{cv}\xi_v + h'_{cv,v'c'}\rho^{(1)}_{v'c'} + h'_{cv,c'v'}\rho^{(1)}_{c'v'} + o_{cv}f(t) && (27)\\ =& \left[(\xi_c - \xi_v)\delta_{cc'}\delta_{vv'} + V_{cv'c'v} - V_{cv'vc'}\right]\rho^{(1)}_{c'v'} + (V_{cc'v'v} - V_{cc'vv'})\rho^{(1)}_{v'c'} + o_{cv}f(t) && (28)\end{aligned}$$

where

$$\begin{aligned} [h'(\rho^0)\cdot\rho^{(1)}]_{mn} =& \sum_{ij} h'_{mn,ij}\rho^{(1)}_{ij}, \quad \frac{\partial h_{mn}}{\partial\rho_{ij}} = V_{mjin} - V_{mjni} && (29)\\ =& \sum_{ij} h'_{mn,ij}\left(\rho^{(1)}_{VC} + \rho^{(1)}_{CV}\right)_{ij} && (30)\\ =& \sum_{v'c'} h'_{mn,v'c'}\rho^{(1)}_{v'c'} + h'_{mn,c'v'}\rho^{(1)}_{c'v'} && (31)\end{aligned}$$

We define the dynamical matrix elements:

$$\mathcal{E}_{cv,c'v'} = (\xi_c - \xi_v)\delta_{cc'}\delta_{vv'} + V_{cv'c'v} - V_{cv'vc'}, \quad \Gamma_{cv,v'c'} = V_{cc'v'v} - V_{cc'vv'} \tag{32}$$

$$\mathcal{E}_{vc,v'c'} = -\mathcal{E}^*_{cv,c'v'}, \quad \Gamma_{vc,c'v'} = -\Gamma^*_{cv,v'c'} \tag{33}$$

To select the causal (retarded) solution, we switch on the perturbation adiabatically from $t = -\infty$: $f(t) \to f(t)\,e^{\eta t}$ with $\eta \to 0^+$. Upon Fourier transform this amounts to the replacement $\omega \to \omega + i\eta$ in every energy denominator. With this prescription understood, the 2 EoMs Eq. (26) and Eq. (28) can be written in the frequency space as:

$$\begin{pmatrix}\hbar\omega & \\ & -\hbar\omega\end{pmatrix}\begin{pmatrix}\rho^{(1)}_{CV}\\ \rho^{(1)}_{VC}\end{pmatrix} = \begin{pmatrix}\mathcal{E} & \Gamma\\ \Gamma^* & \mathcal{E}^*\end{pmatrix}\begin{pmatrix}\rho^{(1)}_{CV}\\ \rho^{(1)}_{VC}\end{pmatrix} + \begin{pmatrix}o_{CV} & \\ & o_{VC}\end{pmatrix} f(\omega) \tag{34}$$

3 Application to Homogeneous Electron Gas

The Hamiltonian of homogeneous electron gas is:

$$\hat{H} = \hat{H}^0 + \hat{V} \tag{35}$$

Where the interaction $\hat{V}$ can be written in (1) fashion:

$$\hat{V} = \frac{1}{2\Omega}\sum_{\mathbf{qkp}}\sum_{\alpha\beta} V(\mathbf{q})\hat{c}^\dagger_{\mathbf{k}\alpha}\hat{c}^\dagger_{\mathbf{p}\beta}\hat{c}_{\mathbf{p+q}\beta}\hat{c}_{\mathbf{k-q}\alpha} = \frac{1}{2}\sum_{ijkl} V_{ijkl}\hat{c}^\dagger_i\hat{c}^\dagger_j\hat{c}_k\hat{c}_l \tag{36}$$

$$V_{ijkl} = \frac{1}{2\Omega}\Big[\underbrace{\delta_{s_is_l}\,\delta_{s_js_k}\,V(\mathbf{p}_i-\mathbf{p}_l)}_{\text{direct}} - \underbrace{\delta_{s_is_k}\,\delta_{s_js_l}\,V(\mathbf{p}_i-\mathbf{p}_k)}_{\text{exchange}}\Big]\,\delta_{\mathbf{p}_i+\mathbf{p}_j,\,\mathbf{p}_k+\mathbf{p}_l} \tag{37}$$

And the kinetic energy term is:

$$\hat{H}^0 = \sum_{\alpha\mathbf{k}}\varepsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha} = \sum_{ij} t_{ij}\hat{c}^\dagger_i\hat{c}_j, \quad t_{ij} = \varepsilon_{\mathbf{p}_i}\delta_{s_is_j}\delta_{\mathbf{p}_i,\mathbf{p}_j} \tag{38}$$

The linear response of density operator is give as:

$$\delta\langle\hat{n}(\mathbf{q})\rangle(\omega) = \sum_{\mathbf{k}\alpha}[\rho^{(1)}_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha)}]_{VC} + [\rho^{(1)}_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha)}]_{CV} \tag{39}$$

$$[\rho^{(1)}_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha)}]_{VC} \propto \Theta(k_F - |\mathbf{k}+\mathbf{q}|)\Theta(|\mathbf{k}|-k_F), \quad [\rho^{(1)}_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha)}]_{CV} \propto \Theta(|\mathbf{k}+\mathbf{q}|-k_F)\Theta(k_F-|\mathbf{k}|) \tag{40}$$

Now we try to explicitly write down the EoM for $\rho^{(1)}_{\mathbf{k+q},\alpha;\mathbf{k}\alpha}$ from Eq. (34):

$$\begin{aligned}\hbar\omega[\rho^{(1)}_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha)}]_{CV} &= \mathcal{E}_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha),(\mathbf{p}\beta)(\mathbf{l}\gamma)}[\rho^{(1)}_{(\mathbf{p}\beta)(\mathbf{l}\gamma)}]_{CV}\\ &+\Gamma_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha),(\mathbf{p}'\beta')(\mathbf{l}'\gamma')}[\rho^{(1)}_{(\mathbf{p}'\beta')(\mathbf{l}'\gamma')}]_{VC} + o_{(\mathbf{k+q}\alpha)(\mathbf{k}\alpha)}f(\omega)\end{aligned} \tag{41}$$

We now evaluate each piece using the HEG interaction matrix element (1). The mapping to (28) is $c = (\mathbf{k}+\mathbf{q},\alpha)$, $v = (\mathbf{k},\alpha)$, $c' = (\mathbf{p},\beta)$, $v' = (\mathbf{l},\gamma)$, with $|\mathbf{k}+\mathbf{q}|, |\mathbf{p}| > k_F$ and $|\mathbf{k}|, |\mathbf{l}| < k_F$.

Term $V_{cv'c'v}$. Setting $i = (\mathbf{k}+\mathbf{q},\alpha)$, $j = (\mathbf{l},\gamma)$, $k = (\mathbf{p},\beta)$, $l = (\mathbf{k},\alpha)$ in the interaction matrix element:

- Momentum conservation: $(\mathbf{k}+\mathbf{q})+\mathbf{l} = \mathbf{p}+\mathbf{k} \;\Rightarrow\; \mathbf{l} = \mathbf{p}-\mathbf{q}$
- Direct: $\delta_{s_is_l}\delta_{s_js_k}V(\mathbf{p}_i-\mathbf{p}_l) = \delta_{\gamma\beta}\,V(\mathbf{q})$
- Exchange: $\delta_{s_is_k}\delta_{s_js_l}V(\mathbf{p}_i-\mathbf{p}_k) = \delta_{\alpha\beta}\delta_{\alpha\gamma}\,V(\mathbf{k}+\mathbf{q}-\mathbf{p})$

$$V_{cv'c'v} = \frac{1}{2\Omega}\big[\delta_{\beta\gamma}\,V(\mathbf{q}) - \delta_{\alpha\beta}\delta_{\alpha\gamma}\,V(\mathbf{k}+\mathbf{q}-\mathbf{p})\big]\,\delta_{\mathbf{l},\,\mathbf{p}-\mathbf{q}} \tag{42}$$

Term $V_{cv'vc'}$. Setting $i = (\mathbf{k}+\mathbf{q}, \alpha)$, $j = (\mathbf{l}, \gamma)$, $k = (\mathbf{k}, \alpha)$, $l = (\mathbf{p}, \beta)$:

- Momentum conservation: same $\delta_{\mathbf{l},\,\mathbf{p}-\mathbf{q}}$
- Direct: $\delta_{s_i s_l}\delta_{s_j s_k} V(\mathbf{p}_i - \mathbf{p}_l) = \delta_{\alpha\beta}\delta_{\gamma\alpha}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p})$
- Exchange: $\delta_{s_i s_k}\delta_{s_j s_l} V(\mathbf{p}_i - \mathbf{p}_k) = \delta_{\gamma\beta}\, V(\mathbf{q})$

$$V_{cv'vc'} = \frac{1}{2\Omega}\big[\delta_{\alpha\beta}\delta_{\alpha\gamma}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p}) - \delta_{\beta\gamma}\, V(\mathbf{q})\big]\ \delta_{\mathbf{l},\,\mathbf{p}-\mathbf{q}} \tag{43}$$

Result. Since terms II and III are exact negatives of each other, they double upon subtraction:

$$V_{cv'c'v} - V_{cv'vc'} = \frac{1}{\Omega}\big[\delta_{\beta\gamma}\, V(\mathbf{q}) - \delta_{\alpha\beta}\delta_{\alpha\gamma}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p})\big]\ \delta_{\mathbf{l},\,\mathbf{p}-\mathbf{q}} \tag{44}$$

and the full dynamical matrix element reads:

$$\begin{aligned}\mathcal{E}_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha),\,(\mathbf{p}\,\beta)(\mathbf{l}\,\gamma)} &= (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})\,\delta_{\mathbf{p},\mathbf{k}+\mathbf{q}}\,\delta_{\mathbf{l},\mathbf{k}}\,\delta_{\alpha\beta}\,\delta_{\alpha\gamma} \\ &\quad + \frac{1}{\Omega}\Big[\underbrace{\delta_{\beta\gamma}\, V(\mathbf{q})}_{\text{direct}} - \underbrace{\delta_{\alpha\beta}\,\delta_{\alpha\gamma}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p})}_{\text{exchange}}\Big]\,\delta_{\mathbf{l},\,\mathbf{p}-\mathbf{q}}\end{aligned} \tag{45}$$

where $\xi_{\mathbf{k}} = \varepsilon_{\mathbf{k}} + \Sigma^{\mathrm{HF}}(\mathbf{k})$ is the Hartree–Fock single-particle energy. Note that:

1. Momentum conservation forces $\mathbf{l} = \mathbf{p} - \mathbf{q}$ in the interaction part, so only particle–hole pairs with the *same* transfer momentum $\mathbf{q}$ couple; different $\mathbf{q}$-sectors decouple.
2. The direct (Hartree) term $\frac{1}{\Omega}\delta_{\beta\gamma}V(\mathbf{q})$ depends only on $\mathbf{q}$, not on $\mathbf{k}$ or $\mathbf{p}$ individually. Retaining only this term (and dropping exchange) recovers **RPA**.
3. The exchange (Fock) term $-\frac{1}{\Omega}\delta_{\alpha\beta}\delta_{\alpha\gamma}V(\mathbf{k}+\mathbf{q}-\mathbf{p})$ depends on the momentum transfer between the two particle–hole pairs and is restricted to same spin ($\alpha = \beta = \gamma$). This is the **TDHF correction beyond RPA**.

Next we evaluate $\Gamma_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha),\,(\mathbf{p}'\,\beta')(\mathbf{l}'\,\gamma')}$ which couples to the VC block $[\rho^{(1)}_{(\mathbf{p}'\,\beta')(\mathbf{l}'\,\gamma')}]_{VC}$ with $|\mathbf{p}'| < k_F$ (valence) and $|\mathbf{l}'| > k_F$ (conduction). From Eq. (28) the mapping is $c = (\mathbf{k}+\mathbf{q}, \alpha)$, $v = (\mathbf{k}, \alpha)$, $v' = (\mathbf{p}', \beta')$, $c' = (\mathbf{l}', \gamma')$, and the definition gives:

$$\Gamma_{cv,\,v'c'} = V_{cc'v'v} - V_{cc'vv'} \tag{46}$$

Term $V_{cc'v'v}$. Setting $i = (\mathbf{k}+\mathbf{q}, \alpha)$, $j = (\mathbf{l}', \gamma')$, $k = (\mathbf{p}', \beta')$, $l = (\mathbf{k}, \alpha)$:

- Momentum conservation: $(\mathbf{k}+\mathbf{q}) + \mathbf{l}' = \mathbf{p}' + \mathbf{k} \;\Rightarrow\; \mathbf{l}' = \mathbf{p}' - \mathbf{q}$
- Direct: $\delta_{s_i s_l}\delta_{s_j s_k}\, V(\mathbf{p}_i - \mathbf{p}_l) = \delta_{\beta'\gamma'}\, V(\mathbf{q})$
- Exchange: $\delta_{s_i s_k}\delta_{s_j s_l}\, V(\mathbf{p}_i - \mathbf{p}_k) = \delta_{\alpha\beta'}\delta_{\alpha\gamma'}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p}')$

$$V_{cc'v'v} = \frac{1}{2\Omega}\big[\delta_{\beta'\gamma'}\, V(\mathbf{q}) - \delta_{\alpha\beta'}\delta_{\alpha\gamma'}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p}')\big]\ \delta_{\mathbf{l}',\,\mathbf{p}'-\mathbf{q}} \tag{47}$$

Term $V_{cc'vv'}$. Setting $i = (\mathbf{k}+\mathbf{q}, \alpha)$, $j = (\mathbf{l}', \gamma')$, $k = (\mathbf{k}, \alpha)$, $l = (\mathbf{p}', \beta')$:

- Momentum conservation: same $\delta_{\mathbf{l}', \mathbf{p}'-\mathbf{q}}$
- Direct: $\delta_{s_i s_l}\delta_{s_j s_k}\, V(\mathbf{p}_i - \mathbf{p}_l) = \delta_{\alpha\beta'}\delta_{\gamma'\alpha}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p}')$
- Exchange: $\delta_{s_i s_k}\delta_{s_j s_l}\, V(\mathbf{p}_i - \mathbf{p}_k) = \delta_{\gamma'\beta'}\, V(\mathbf{q})$

$$V_{cc'vv'} = \frac{1}{2\Omega}\left[\delta_{\alpha\beta'}\delta_{\alpha\gamma'}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p}') - \delta_{\beta'\gamma'}\, V(\mathbf{q})\right]\, \delta_{\mathbf{l}', \mathbf{p}'-\mathbf{q}} \tag{48}$$

Result. Again the two terms are exact negatives, so:

$$\Gamma_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha),\,(\mathbf{p}'\,\beta')(\mathbf{l}'\,\gamma')} = \frac{1}{\Omega}\Big[\underbrace{\delta_{\beta'\gamma'}\, V(\mathbf{q})}_{\text{direct}} - \underbrace{\delta_{\alpha\beta'}\,\delta_{\alpha\gamma'}\, V(\mathbf{k}+\mathbf{q}-\mathbf{p}')}_{\text{exchange}}\Big]\, \delta_{\mathbf{l}', \mathbf{p}'-\mathbf{q}} \tag{49}$$

Note that Γ has exactly the same interaction structure as the off-diagonal part of $\mathcal{E}$ in (45) (with the replacement $\mathbf{p} \to \mathbf{p}'$, $\beta \to \beta'$, $\gamma \to \gamma'$), but without the diagonal quasiparticle energy term. Momentum conservation again forces $\mathbf{l}' = \mathbf{p}' - \mathbf{q}$, confirming that the $\mathbf{q}$-sector decoupling holds for the full EoM including the Γ coupling between CV and VC blocks.

3.1 RPA equation of motion

Dropping the exchange terms in (45) and (49), the dynamical matrices reduce to:

$$\mathcal{E}^{\text{RPA}}_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha),\,(\mathbf{p}\,\beta)(\mathbf{l}\,\gamma)} = (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})\, \delta_{\mathbf{p},\mathbf{k}+\mathbf{q}}\, \delta_{\mathbf{l},\mathbf{k}}\, \delta_{\alpha\beta}\, \delta_{\alpha\gamma} + \frac{1}{\Omega}\, \delta_{\beta\gamma}\, V(\mathbf{q})\, \delta_{\mathbf{l}, \mathbf{p}-\mathbf{q}} \tag{50}$$

$$\Gamma^{\text{RPA}}_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha),\,(\mathbf{p}'\,\beta')(\mathbf{l}'\,\gamma')} = \frac{1}{\Omega}\, \delta_{\beta'\gamma'}\, V(\mathbf{q})\, \delta_{\mathbf{l}', \mathbf{p}'-\mathbf{q}} \tag{51}$$

Substituting into Eq. (41) and performing the contractions (summing over $\mathbf{p}, \beta, \mathbf{l}, \gamma$ and $\mathbf{p}', \beta', \mathbf{l}', \gamma'$):

$\mathcal{E}$ term. The diagonal part collapses via the Kronecker deltas:

$$(\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})\, \delta_{\mathbf{p},\mathbf{k}+\mathbf{q}}\, \delta_{\mathbf{l},\mathbf{k}}\, \delta_{\alpha\beta}\, \delta_{\alpha\gamma}\, [\rho^{(1)}_{(\mathbf{p}\,\beta)(\mathbf{l}\,\gamma)}]_{CV} = (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})\, [\rho^{(1)}_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha)}]_{CV} \tag{52}$$

The direct part, after eliminating $\mathbf{l} = \mathbf{p} - \mathbf{q}$ and $\beta = \gamma$:

$$\frac{V(\mathbf{q})}{\Omega} \sum_{\mathbf{p}\,\beta} [\rho^{(1)}_{(\mathbf{p}\,\beta)(\mathbf{p}-\mathbf{q}\,\beta)}]_{CV} \tag{53}$$

Relabelling $\mathbf{p} = \mathbf{k}' + \mathbf{q}$ (with $|\mathbf{k}'+\mathbf{q}| > k_F$, $|\mathbf{k}'| < k_F$), this becomes:

$$\frac{V(\mathbf{q})}{\Omega} \sum_{\mathbf{k}'\,\beta} [\rho^{(1)}_{(\mathbf{k}'+\mathbf{q}\,\beta)(\mathbf{k}'\,\beta)}]_{CV} \tag{54}$$

Γ term. After eliminating $\mathbf{l}' = \mathbf{p}' - \mathbf{q}$ and $\beta' = \gamma'$:

$$\frac{V(\mathbf{q})}{\Omega} \sum_{\mathbf{p}'\,\beta'} [\rho^{(1)}_{(\mathbf{p}'\,\beta')(\mathbf{p}'-\mathbf{q}\,\beta')}]_{VC} \tag{55}$$

Relabelling $\mathbf{k}' = \mathbf{p}' - \mathbf{q}$ (so $\mathbf{p}' = \mathbf{k}' + \mathbf{q}$, with $|\mathbf{k}'+\mathbf{q}| < k_F$ and $|\mathbf{k}'| > k_F$):

$$\frac{V(\mathbf{q})}{\Omega} \sum_{\mathbf{k}'\,\beta'} [\rho^{(1)}_{(\mathbf{k}'+\mathbf{q}\,\beta')(\mathbf{k}'\,\beta')}]_{VC} \tag{56}$$

Combined RPA EoM. Recognising that the two interaction sums together reconstruct the density response from Eq. (41):

$$\delta\langle\hat{n}(\mathbf{q})\rangle(\omega) = \sum_{\mathbf{k}'\,\beta}\left([\rho^{(1)}_{(\mathbf{k}'+\mathbf{q}\,\beta)(\mathbf{k}'\,\beta)}]_{CV} + [\rho^{(1)}_{(\mathbf{k}'+\mathbf{q}\,\beta)(\mathbf{k}'\,\beta)}]_{VC}\right) \tag{57}$$

the RPA equation of motion becomes:

$$\boxed{\hbar\omega\,[\rho^{(1)}_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha)}]_{CV} = (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})\,[\rho^{(1)}_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha)}]_{CV} + \frac{V(\mathbf{q})}{\Omega}\,\delta\langle\hat{n}(\mathbf{q})\rangle(\omega) + o_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha)}\,f(\omega)} \tag{58}$$

The direct interaction term depends on $\mathbf{k}$ and α only through the left-hand side; the coupling to all other particle–hole pairs is entirely through the collective density fluctuation $\delta\langle\hat{n}(\mathbf{q})\rangle(\omega)$, which is the hallmark of RPA screening. We can obtain the other half of Eq. (34) through complex conjugation. First the Hermitian nature of ρ implies:

$$\rho^{(1)}_{vc}(t) = (\rho^{(1)}_{cv}(t))^* \quad\Rightarrow\quad \rho^{(1)}_{vc}(\omega) = (\rho^{(1)}_{cv}(-\omega))^* \tag{59}$$

complex conjugating the CV block EoM Eq. (58) and replacing $\omega \to -\omega$ gives the VC block EoM:

$$-\hbar\omega[\rho^{(1)}_{(\mathbf{k}\alpha)(\mathbf{k}+\mathbf{q}\alpha)}]_{VC} = (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})[\rho^{(1)}_{(\mathbf{k}\alpha)(\mathbf{k}+\mathbf{q}\alpha)}]_{VC} + \frac{V(\mathbf{q})}{\Omega}[\delta\langle\hat{n}(\mathbf{q})\rangle(-\omega)]^* + o_{(\mathbf{k}\alpha)(\mathbf{k}+\mathbf{q}\alpha)}f(-\omega)^* \tag{60}$$

$$= (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})[\rho^{(1)}_{(\mathbf{k}\alpha)(\mathbf{k}+\mathbf{q}\alpha)}]_{VC} + \frac{V(\mathbf{q})}{\Omega}\delta\langle\hat{n}(-\mathbf{q})\rangle(\omega) + o_{(\mathbf{k}\alpha)(\mathbf{k}+\mathbf{q}\alpha)}f(\omega) \tag{61}$$

To summarise, $\rho^{(1)}_{CV}$ and $\rho^{(1)}_{VC}$ are solved as:

$$\boxed{[\rho^{(1)}_{(\mathbf{k}+\mathbf{q}\alpha)(\mathbf{k}\alpha)}]_{CV} = \frac{\frac{V(\mathbf{q})}{\Omega}\delta\langle\hat{n}(\mathbf{q})\rangle(\omega) + o_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha)}\,f(\omega)}{\hbar(\omega+i\eta) - (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})}(1 - f_{\mathbf{k}+\mathbf{q}})f_{\mathbf{k}}} \tag{62}$$

$$\boxed{[\rho^{(1)}_{(\mathbf{k}+\mathbf{q}\alpha)(\mathbf{k}\alpha)}]_{VC} = \frac{\frac{V(\mathbf{q})}{\Omega}\delta\langle\hat{n}(\mathbf{q})\rangle(\omega) + o_{(\mathbf{k}+\mathbf{q}\,\alpha)(\mathbf{k}\,\alpha)}\,f(\omega)}{-\hbar(\omega+i\eta) - (\xi_{\mathbf{k}} - \xi_{\mathbf{k}+\mathbf{q}})}f_{\mathbf{k}+\mathbf{q}}(1 - f_{\mathbf{k}})} \tag{63}$$

Where we abbreviate the Fermi occupation factors as $f_{\mathbf{k}} = \Theta(\varepsilon_F - \xi_{\mathbf{k}})$. Usually the external field is independent of the system's internal momenta, therefore we here take $o_{(\mathbf{k}+\mathbf{q}\alpha)(\mathbf{k}\alpha)} = o_{\mathbf{q}}$ and Eq. (57) can be solved as:

$$\Pi^0(\mathbf{q},\omega) = \sum_{\mathbf{k}}\frac{(1 - f_{\mathbf{k}+\mathbf{q}})f_{\mathbf{k}} - f_{\mathbf{k}+\mathbf{q}}(1 - f_{\mathbf{k}})}{\hbar(\omega+i\eta) - (\xi_{\mathbf{k}+\mathbf{q}} - \xi_{\mathbf{k}})} \tag{64}$$

$$\delta\langle\hat{n}(\mathbf{q})\rangle(\omega) = \Pi^0(\mathbf{q},\omega)\left(\frac{V(\mathbf{q})}{\Omega}\delta\langle\hat{n}(\mathbf{q})\rangle(\omega) + o_{\mathbf{q}}f(\omega)\right) \tag{65}$$

$$\Rightarrow \delta\langle\hat{n}(\mathbf{q})\rangle(\omega) = \frac{\Pi^0(\mathbf{q},\omega)}{1 - \frac{V(\mathbf{q})}{\Omega}\Pi^0(\mathbf{q},\omega)}o_{\mathbf{q}}f(\omega) \tag{66}$$

The $i\eta$ prescription makes Π^0 the *retarded* response function, whose poles lie below the real ω-axis, ensuring causality. $\Pi^0(\mathbf{q},\omega)$ has the exact form of the Lindhard function [2], except that the energy difference in the denominator is from the Hartree-Fock single-particle spectrum instead of the non-interacting one:

$$\xi_{\mathbf{k}\alpha} = \varepsilon_{\mathbf{k}} + \frac{1}{\Omega}\sum_{\mathbf{q}\gamma}[V(0) - \delta_{\alpha\gamma}V(\mathbf{k}-\mathbf{q})]\,\rho^0_{\mathbf{q}\gamma}, \quad \rho^0_{\mathbf{q}\gamma} = \langle\hat{c}^\dagger_{\mathbf{q}\gamma}\hat{c}_{\mathbf{q}\gamma}\rangle_0 \tag{67}$$

The diverging $V(0)$ is cancelled by the positive background charge, and if we ignore the remaining exchagne term, we recover the original RPA density-density response function.

References

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[2] A. L.Fetter and J. D. Walecka, *Quantum Theory of Many-Particle Systems* (McGraw-Hill, New York, 1971).